



ArticulateX: End-to-End Monolingual Speech Translation in Articulator Space

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Abstract

We present ArticulateX, the first non-autoregressive direct speech-to-speech translation (S2ST) model that operates through an articulatory latent space, offering an efficient alternative to existing cascaded models. It consists of a direct speech-to-articulator encoder, a latent articulator-to-MelSpectrogram mapper, and a vocoder for high-fidelity speech synthesis. By leveraging articulatory representations, which are inherently language-agnostic, our model effectively captures speech dynamics, preserving speaker identity, prosody and expressiveness across languages. Unlike prior autoregressive models, ArticulateX eliminates the need for intermediate text, discrete units and/or complex self-supervised objectives, enabling faster inference, stable training, and improved translation quality. We demonstrate the efficacy of the proposed model in fr-en and de-en speech-to-speech translation on the CVSS dataset, achieving BLEU scores better or comparable to existing models.

Index Terms: Speech translation, neural networks, end-to-end modeling

1. Introduction

During the past decades, speech-to-speech translation (S2ST) [1] has emerged as a critical tool for real-time cross-linguistic communication. Traditional S2ST systems are based on modular pipelines, typically involving Automatic Speech Recognition (ASR) [2], Machine Translation (MT) [3], and Text-to-Speech (TTS) [4]. Although effective, these systems suffer from significant drawbacks, including latency, compound errors, and reliance on language-specific intermediate representations. Recent advances in machine learning, such as sequence-to-sequence models [5] and self-supervised learning [6], have enabled end-to-end S2ST systems that bypass textual representations, introducing new possibilities for efficiency and scalability. Models like Translatotron [7, 8] and UnitY [9] have demonstrated the potential of language-agnostic acoustic units and multilingual pre-training. However, aligning disparate acoustic features such as phonology, prosody, and rhythm remains a significant challenge for direct speech-to-speech systems.

In this paper, we present *ArticulateX*, a novel non-autoregressive framework [10] that introduces a *language-independent articulatory latent space* for direct speech-to-speech translation. Unlike conventional acoustic or spectral representations, our approach maps speech signals to dynamic articulatory gesture configurations of the vocal tract that underlie speech production across all languages. Drawing inspiration from articulatory phonetics and the motor theory of speech perception [11], ArticulateX models speech as a universal set of ar-

ticulatory movements, abstracting away language and speaker-specific variability. This enables more effective cross-linguistic alignment and addresses challenges related to acoustic variability and coarticulatory effects.

The ArticulateX framework¹ employs an end-to-end design that consists of three key components: a source audio to target articulator encoder to map input speech into articulatory representations, a latent articulator-to-MelSpectrogram generator to generate target speech features, and a vocoder for high-quality speech synthesis. Using state-of-the-art deep learning architectures, including convolutional layers, conformer blocks [12], and transformer [13], our model captures subtle articulatory dynamics essential for natural synthesis while remaining robust to speaker variability and environmental noise. Furthermore, the non-autoregressive architecture enables parallelized inference, significantly reducing latency compared to autoregressive models. By introducing articulatory representations as a universal intermediary, ArticulateX provides a scalable, efficient, and robust solution for multilingual speech-to-speech translation. Experimental results demonstrate its ability to achieve high translation accuracy and speech naturalness while addressing the inherent limitations of existing S2ST systems².

In summary, this work makes the following contributions.

1. We propose a novel speech-to-speech translation framework that takes advantage of a language-agnostic articulatory space to bridge the gap between source and target languages.
2. We present a detailed methodology for mapping acoustic signals to articulatory representations using deep neural networks.
3. We provide extensive experimental evaluations that highlight the robustness of our system.

2. Related Works

Conventional S2S translation systems are implemented as a cascade of three pre-trained components: automatic speech recognition (ASR) for the source language, machine translation (MT) for the conversion of the source language to the target language, and text-to-speech synthesis (TTS) for speech generation in the target language [1]. Recent works have reduced the dependence on multiple components by utilising learned discrete speech representations as an intermediate representation. For instance, recently, Zhang et al. [14] jointly trained a phoneme-conditioned VAE model for a multilingual setting, while Lee et al. [15] used a pre-trained vocoder to directly predict the waveform from the discrete representation without re-

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²<https://sites.google.com/view/articulatex-demo/home>

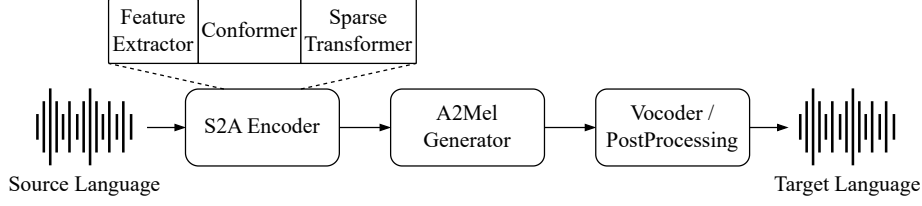


Figure 1: *ArticulateX: Translates speech from source to English language using latent articulatory representations of the vocal tract.*

lying on spectrograms. In addition, there have been attempts to enhance the linguistic information in the latent space using language-aware normalisation [16]. However, end-to-end training at scale is often infeasible, which may limit their overall performance. Further, these approaches are not as flexible, as they cannot easily adapt to new or low-resource languages. End-to-end S2S translation systems have gained considerable attention with the Translatotron models from Google [17, 7, 8]. Similarly, work in [18] proposes a simple yet effective framework for multilingual end-to-end S2S. Recent advancements in self-supervised learning (SSL) based models (e.g., wav2vec or HuBERT) pre-trained on large amount of unlabeled audio data have gained traction for extracting linguistically rich acoustic features. For S2ST task, these models enabled cross-lingual transfer; for instance, XLS-R [19], a multilingual model pre-trained on 436K hours of speech, achieved few-shot translation for under-resourced languages like Swahili by aligning latent speech representations across languages. Alternatively, some studies employ discrete acoustic units as an intermediate representation to bridge the speech-text modality gap. For instance, the UnitY framework [9] converts source speech into language-agnostic units via HuBERT and then generates target speech using a multilingual TTS module, reducing training data needs by 60% compared to spectrogram-based systems while maintaining high naturalness (MOS). Additionally, non-autoregressive models like FastSpeech-2 [4] achieve sub-second latency through parallel spectrogram prediction, and other works have enhanced synthesis quality by preserving speaker traits [20].

Existing models exhibit several limitations: (1) robustness challenges, including over-/under-generation and difficulties with long sequences; (2) reliance on autoregressive architectures that yield slow inference and degrade quality when replaced by non-autoregressive synthesis; and (3) closed-source, data-intensive, and expensive training processes that hinder open-source progress.

3. ArticulateX

In this section, we introduce ArticulateX, which bypasses textual intermediaries by directly mapping acoustic signals into a language-agnostic articulatory domain. The pipeline comprises of three modules: (1) S2A Encoder, (2) MelSpectrogram Generator and (3) Vocoder/PostProcessing.

1. **Speech-2-Articulator (S2A) Encoder:** is a core innovation in ArticulateX, mapping raw speech from the source language to a language-independent articulatory latent space of the target language. The S2A Encoder integrates a feature extractor, Conformer blocks, and a sparse transformer, all carefully optimized to balance expressive power, efficiency, and scalability. By operating in a universal articulatory space, the S2A Encoder eliminates language-specific biases,

making it adept at modeling cross-lingual speech dynamics. This ensures robust and natural translations, even for low-resource languages. Inspired by LoRA [21], all trainable layers are reparameterized into low-rank representations (rank 4), significantly reducing computational overhead and model size (3.1M parameters) while retaining expressiveness. This makes ArticulateX scalable and efficient for real-world applications. Figure 2(a), 2(c), and 2(b) depict the components of the S2A encoder.

- **Feature Extractor:** employs four dilated 1D-convolutional blocks (Conv-BN-LeakyReLU) with progressively larger kernel sizes & dilation factors to capture long-term temporal dependencies in speech. This is followed by four standard Conv1D layers with increasing kernel size & strides to progressively downsample features, culminating in an attentive average pooling layer that emphasizes the most informative temporal segments.
- **Conformer Block:** The four Conformer blocks [12] seamlessly enable the model to capture both local acoustic nuances and global speech patterns, crucial for preserving speaker characteristics and prosody in the translation process.
- **Sparse Transformer:** is a lightweight variant of the standard encoder-decoder transformer [13] with dense layers replaced with sparse linear layers that use a trainable softmax threshold. To stabilize training, a Straight-Through Estimator (STE) is applied during backpropagation, ensuring gradients remain smooth and effective while training with sparse inputs/outputs.

Articulatory representations encode both local dependencies (e.g., transitions between the neighbouring phonemes) and global dynamics (e.g., rhythm, prosody, or speaker-level characteristics). A Conformer block is explicitly designed to handle this dual nature by combining convolutions for local modeling with self-attention for global relationships. Further, articulatory representations are highly sparse in nature, meaning that only a subset of features is active at a given time. Thus, a sparse transformer is well-suited for such data as it can selectively attend to the most salient regions in the feature space, avoiding overfitting on inactive or less relevant dimensions of the articulatory representation. A large Transformer alone excel at capturing global patterns but lacks the inductive bias to efficiently model fine-grained local details, which are critical for articulatory mapping.

2. **Articulator to MelSpectrogram (A2Mel) Generator:** in ArticulateX transforms sparse articulatory feature sequences in the target language into high-fidelity MelSpectrograms. It addresses the unique challenges of mapping sparse representations to dense spectrograms while preserving fine-grained prosodic and acoustic details. Our A2Mel Generator em-

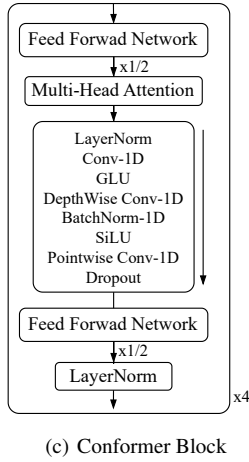
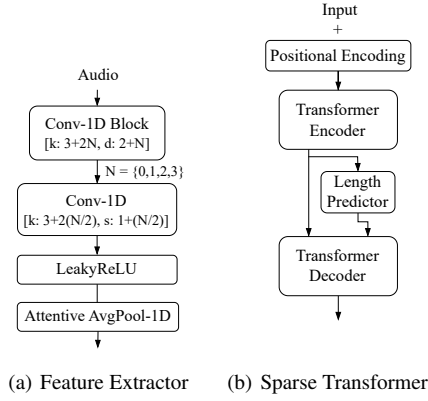


Figure 2: *Components of S2A Encoder. k , d , s denotes Kernel-Size, Dialation and Stride, respectively*

plays a lightweight FastSpeech-2-inspired architecture (50M parameters) optimized for articulatory input, integrating pitch and energy conditioning for every articulatory unit, ensuring language-agnostic fine-grained prosody control. Additionally, instead of relying on a separate aligner, the system incorporates a self-supervised duration predictor trained jointly with the generator. This duration predictor effectively aligns sparse articulatory features with their temporal representations, eliminating the need for external alignment tools while improving training efficiency.

3. **Vocoder/PostProcessing:** We employ a pretrained HiFi-GAN-V1 [22] (with and without fine-tuning; 14M parameters) to transform the spectrograms into raw waveforms. To improve details in high frequencies and remove temporal inconsistencies, we used the CDiffuse model (4M parameters) from [23].

4. Experimental setup

4.1. Dataset

We conducted experiments using the benchmark CVSS Corpus. The dataset has 1,900 hours of sentence-level parallel speech from 21 source languages to English generated via high-quality

Table 1: *S2S Translation performance comparison using BLEU Scores for various systems. Here, ‘Mono’ & ‘Multi’ denote mono-lingual & multi-lingual models, respectively.*³

Model	Params (Approx.)	BLEU	
		fr-en	de-en
Translatotron-2 [7] [Multi]	-	26.13	16.92
UnitY [9] [Multi]	-	26.9	16.36
S2UT [15] [Mono]	130 M	22.23	2.99
DASpeech [26] [Mono]	95 M	25.03	-
ComSpeech [27] [Multi]	112 M	28.15	18.16
Hibiki [28] [Mono]	2.7 B	38.21	-
Ours [Mono]	70 M	37.24	20.07

TTS models. It has two variants: CVSS-C, offering synthesized speech in a canonical voice, and CVSS-T, with the source speaker’s voice preserved. This work focuses on the former, specifically fr-en and de-en pairs.

4.2. Model Training

Our overall pipeline is implemented in PyTorch and is architecture-agnostic for constituent building blocks. First, the target language text is converted into a sequence of phonemes that are then converted to binary-encoded articulatory features. S2A encoder is trained using a composite loss that combines masked prediction loss for accurate sequence length estimation and MAE loss for precise articulator feature reconstruction. For extracting articulatory features and the A2Mel generator, we adapted the implementation and training pipeline from the IMS Toucan toolkit [24]. Finally, the HiFiGAN vocoder is fine-tuned using a composite loss on the generated raw waveforms. The first loss is the standard MelLoss from [22], while the second one is a multi-resolution STFT loss proposed in [25].

Our model components are trained on a machine with $3 \times$ NVIDIA RTX 3090 GPUs (24 GB each) and 512 GB of RAM. Model optimization is achieved with the Adam optimizer, batchsize of 64 and a learning rate of 10^{-3} with a cosine annealing scheduler. The STE in sparse transformer also uses gradient clipping. Our S2A encoder is primarily trained on 4-sec audio segments and, together with A2Mel+Vocoder ArticulateX can successfully translate sequences up to 12 seconds.

5. Results

In this section, we present an evaluation of our proposed system ArticulateX for fr-en & de-en translation. We compare its performance against six SOTA models: Translatotron-2, UnitY, S2UT, DASpeech, ComSpeech and Hibiki.

5.1. Comparison with existing systems

The performance in terms of BLEU scores (higher the better) of the proposed and existing models for speech translation is summarized in Table 1. To calculate the BLEU score, we employ the pretrained *Wav2Vec2-vox-960h* ASR model on the respective CVSS testset.

Notably, our proposed model, with a minimal parameter count of approximately 70M, achieves BLEU scores of 37.24 fr-en and 20.07 de-en, which are competitive with or superior to those obtained by considerably larger translation models. For instance, compared to the recent Hibiki model, ArticulateX achieves only 2.4% relative lower performance on

³The number of parameters in Translatotron-2 and UnitY models is unknown.

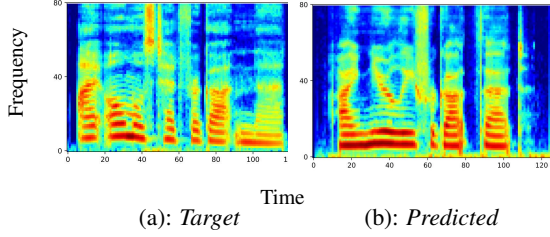


Figure 3: *Target and Predicted Spectrograms for (de: Ich hätte das fast vergessen.), (Target en: I almost forgot that), (Predicted en: I nearly forgot that)*

(fr-en). In most existing works, both models, such as the latent representation-based Translatron-2 model or latent discrete unit-based DASpeech, UnitY & S2UT models, the training phase arguments the acoustic model with auxiliary information via additional linguistic decoding tasks. In contrast, ArticulateX doesn’t rely on any separate linguistic information and consists of modular & exchangeable blocks that can be pre-trained or trained in an end-to-end manner. Alternatively, recent works such as Translatron-3 and ComSpeech have also explored unsupervised training without parallel translation data. In a preliminary comparison with ComSpeech, we observe that supervised models like ArticulateX or Hibiki have far superior performance. These findings demonstrate the efficacy of our lightweight architecture in achieving high translation quality, thereby highlighting a favourable trade-off between model complexity and performance.

5.2. Qualitative Evaluation & Ablation Study

To evaluate the naturalness and equivalency of the translated outputs, we designed a rigorous Mean Opinion Score (MOS) experiment involving English-speaking individuals with English as their second language. Each participant was presented with a randomized set of translation samples to mitigate potential biases and was instructed to evaluate the outputs based on two key aspects: 1) naturalness, which measures the human-likeness of the translation, and 2) equivalency, which assesses the accuracy of the meaning relative to the source text. Raters used a standardized Likert scale ranging from 1 (poor) to 5 (excellent), with comprehensive scoring guidelines provided to ensure consistency across evaluations. By averaging the scores across all samples and participants, we derived an aggregate MOS for our system.

In order to ascertain the contribution of individual components to our system’s overall performance, we conducted an extensive ablation study. Specifically, we explain below a few approaches that resulted in intelligible outputs.

1. Initially, we attempted to map the articulatory representations of the source language (*fr/de*) to those of the target language using classical machine learning techniques. We explored various configurations combining Gaussian mixture models (GMM) and support vector regression (SVR). Specifically, a GMM was trained on unique articulatory representations to predict posterior probabilities for each articulator, while a multivariate SVR was simultaneously trained on the same data. The outputs of both models were combined via a weighted average to obtain the final articulatory representation, which was then input to a pretrained A2Mel+Vocoder

Table 2: *Qualitative and MOS performance of speech-2-articulatory (S2A), articulatory-2-articulatory (A2A) and text-2-articulatory (T2A) mapping. ArticulateX_V1 includes the post-processing network.*

Method	Approach	Naturalness	Equivalence
A2A	SVR + GMM	2.04 ± 0.04	1.89 ± 0.04
T2A	Conv1D	2.56 ± 0.04	2.01 ± 0.04
	Transformer	3.24 ± 0.05	2.66 ± 0.05
	Sparse Transformer	1.97 ± 0.04	2.99 ± 0.07
S2A	ArticulateX	4.79 ± 0.01	4.01 ± 0.01
	ArticulateX_V1	4.84 ± 0.01	4.24 ± 0.01

model for audio synthesis. Although this approach produced approximate mappings, it was effective only for very short durations, yielding suboptimal translation for longer sequences.

2. In our second ablation study, we directly mapped source language text to the target language articulator space using three sequence models: a Conv1D-based model, a Transformer, and a Sparse Transformer. Here, text embeddings were extracted using Bert-based MiniLM [29]. The Conv1D model comprises 12 Conv1D–BatchNorm–ReLU blocks with residual connections every 3rd block, followed by a softmax layer ($\approx 11M$ parameters). Both the Transformer and Sparse Transformer follow the architecture of [13] with three encoder and three decoder layers; the Sparse Transformer replaces standard linear layers with custom sparse variants, containing 22M and 10M parameters, respectively. The resulting articulatory representations are synthesized into audio via a pre-trained A2Mel+Vocoder. Notably, the Conv1D and Transformer models yield greater naturalness, while the Sparse Transformer offers superior equivalence.
3. Finally, the third set of experiments involves our proposed direct speech-to-articulatory mapping approach. Here, we experimented with the base ArticulateX model, and ArticulateX_V1 includes the post-processing CDiffuse model.

The results of these experiments are reported in Table 2. The comparative analysis clearly indicates that our S2A-based approach in ArticulateX delivers significantly higher MOS scores for both naturalness and equivalence compared to A2A & T2A approaches. Additionally, our system achieves an RTF of 0.97, demonstrating near real-time processing capability. These results underscore the efficacy of audio-based articulatory representations in improving the quality and fidelity of speech synthesis, and they provide a robust foundation for future developments in speech translation technologies.

6. Conclusion & Future Work

We presented ArticulateX, a non-autoregressive direct S2S translation model that innovatively leverages a language-independent articulatory latent space for efficient and high-fidelity translation. With its novel lightweight S2A Encoder and A2Mel Generator, ArticulateX achieves state-of-the-art performance by combining sparse articulatory representations with computationally efficient modules like Conformer blocks and Sparse Transformers. Future work will extend the training dataset to include longer sequences for improved temporal modeling and explore multilingual capabilities through architectural and training paradigm enhancements, aiming to make ArticulateX a robust, scalable solution for S2S translation.

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